Introduction to statistics

(Day 3)

Recap

Recap

- Statistics is divided in *Descriptive* and *Inferential*

- Categorical variables are described with absolute and relative frequencies, numerical ones are described with measures of central tendency, dispersion, and correlation
- Variables can be summarised with multiple graphical representations
- Parameters (calculated on the population) *vs* statistics (calculated on the sample)

Recap

- The Normal distribution allows us to determine both the probability of observing a certain value and its expected frequency
- The z-score allows us to position an observation relative to the reference population and to compare data from very different distributions

- We use statistics to estimate parameters (point estimates), which should be communicated along with a measure of confidence (CI or ME; interval estimates)
- A 95% confidence interval results from a procedure that, in 95% of cases, contains the true population value.

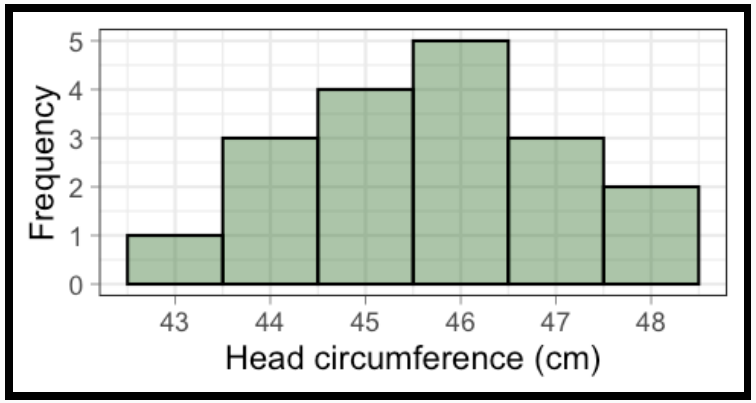
Making decision with data

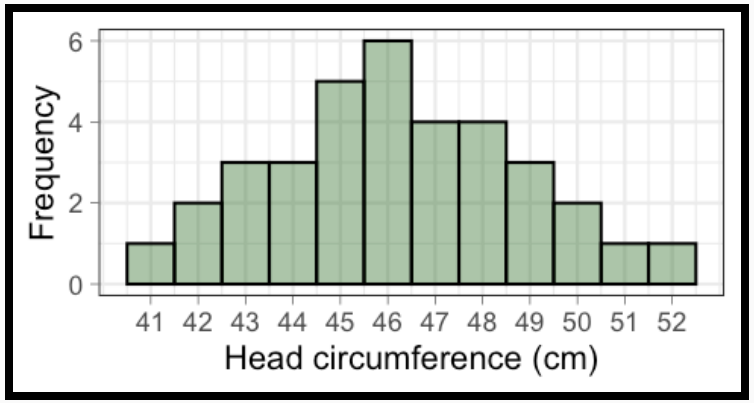
Learning objectives

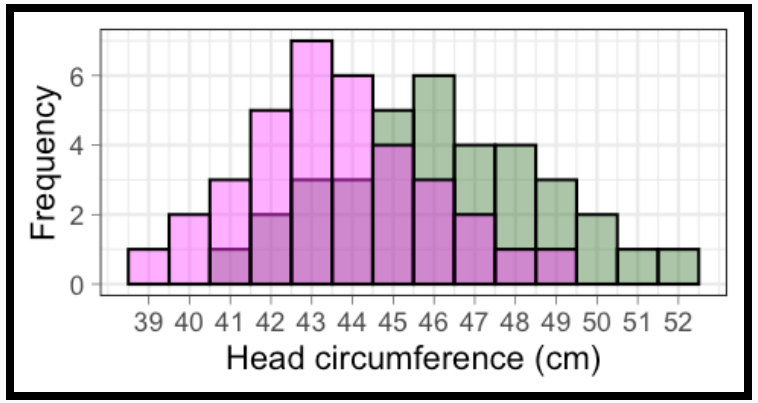
- Make and test hypotheses
- Interpret P values
- Distinguish between Type I and Type II errors
- Understand the importance of the statistical power

Why we observe data variability ?

- We will see how to determine whether the variability depends on the experimental conditions or whether it is due to individual differences and/or measurement errors.







⚠️ Disclaimer ⚠️

This part continues to be complex.
Don’t worry, we're almost there!

Making hypotheses

> *A hypothesis can be defined as a proposed explanation for a phenomenon. It is not the absolute truth, but a provisional, working assumption, perhaps best thought of as a potential suspect in a criminal case.*

Spiegelhalter, D., *The Art of Statistics: Learning From Data*, Pelican, 2019

Making hypotheses

- The outcomes in the intervention and the control group are different
- The proportion of an event in the intervention and control group is different

Exercise #1

? Suppose our hypothesis is that everyone who lives beyond 90 years of age is a non-smoker

To investigate this hypothesis, it is easier to:

a) <u>Prove the hypothesis</u> by finding every single person aged 90 or older and verifying that all of them are non-smokers
b) <u>Refute the hypothesis</u> by finding just one person aged 90 or older who is a smoker

The falsification principle and the null hypothesis

- The outcomes in the intervention and the control group are ~~different~~ **the same**
- The proportion of an event in the intervention and control group is ~~different~~ **the same**

Exercise #2

Objective To determine whether intravenous dexamethasone increases the number of ventilator-free days among patients with COVID-19–associated ARDS.

Design, Setting, and Participants Multicenter, randomized, open-label, clinical trial conducted in 41 intensive care units (ICUs) in Brazil. Patients with COVID-19 and moderate to severe ARDS, according to the Berlin definition, were enrolled from April 17 to June 23, 2020. Final follow-up was completed on July 21, 2020. The trial was stopped early following publication of a related study before reaching the planned sample size of 350 patients.

? Which is the null hypothesis of this study?

a) Dexamethasone plus standard care is **more effective** than standard care alone
b) Dexamethasone plus standard care is **less effective** than standard care alone
c) Dexamethasone plus standard care is **as effective** as standard care alone
d) Dexamethasone plus standard care is not a **as effective** as standard care alone

Tomazini, B.M., *et al.*, Effect of dexamethasone on days alive and ventilator-free in patients with moderate or severe acute respiratory distress syndrome and COVID-19: the CoDEX randomized clinical trial.", JAMA, 2020, doi:10.1001/jama.2020.17021

Exercise #3

Objective To determine whether intravenous dexamethasone increases the number of ventilator-free days among patients with COVID-19–associated ARDS.

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? How do you define the null hypothesis in this study?

a) $\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$
b) $\mu_{\mathrm{c}} - \mu_{\mathrm{i}} \neq 0$
c) $\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}} = 0$
d) $\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}} \neq 0$

Making hypotheses

📌 Dexamethasone plus standard care is **as effective as** standard care

$$n_{\mathrm{i}} = 151, \bar{x}_{\mathrm{i}} = 6.6, s_{\mathrm{i}} = 10.0$$
$$n_{\mathrm{c}} = 148, \bar{x}_{\mathrm{c}} = 4.0, s_{\mathrm{c}} = 8.7$$

$$\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$$
$$\rightarrow \text{Null hypothesis } (\mathcal{H}_0)$$

Making hypotheses

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$$\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$$
$$\rightarrow \text{Null hypothesis } (\mathcal{H}_0)$$

$$\mu_{\mathrm{c}} - \mu_{\mathrm{i}} \neq 0$$
$$\rightarrow \text{Alternative hypothesis } (\mathcal{H}_1/\mathcal{H}_A)$$

Exercise #4

? The fact that the null hypothesis and the alternative hypothesis are mutually exclusive means that if the null hypothesis is true, the alternative hypothesis

a) must also be true
b) may be either true or false
c) must be false
d) depends on the alternative hypothesis

Testing hypotheses

📌 Dexamethasone plus standard care is **as effective as** standard care

Interventions Twenty mg of dexamethasone intravenously daily for 5 days, 10 mg of dexamethasone daily for 5 days or until ICU discharge, plus standard care (n=151) or standard care alone (n=148).

Results A total of 299 patients (mean [SD] age, 61 [14] years; 37% women) were enrolled and all completed follow-up. Patients randomized to the dexamethasone group had a mean 6.6 ventilator-free days (95% CI, 5.0-8.2) during the first 28 days vs 4.0 ventilator-free days (95% CI, 2.9-5.4) in the standard care group

$$n_\mathrm{i} = 151, \quad \bar{x}_\mathrm{i} = 6.6, \quad s_\mathrm{i} = 10.0$$
$$n_\mathrm{c} = 148, \quad \bar{x}_\mathrm{c} = 4.0, \quad s_\mathrm{c} = 8.7$$

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Testing hypotheses

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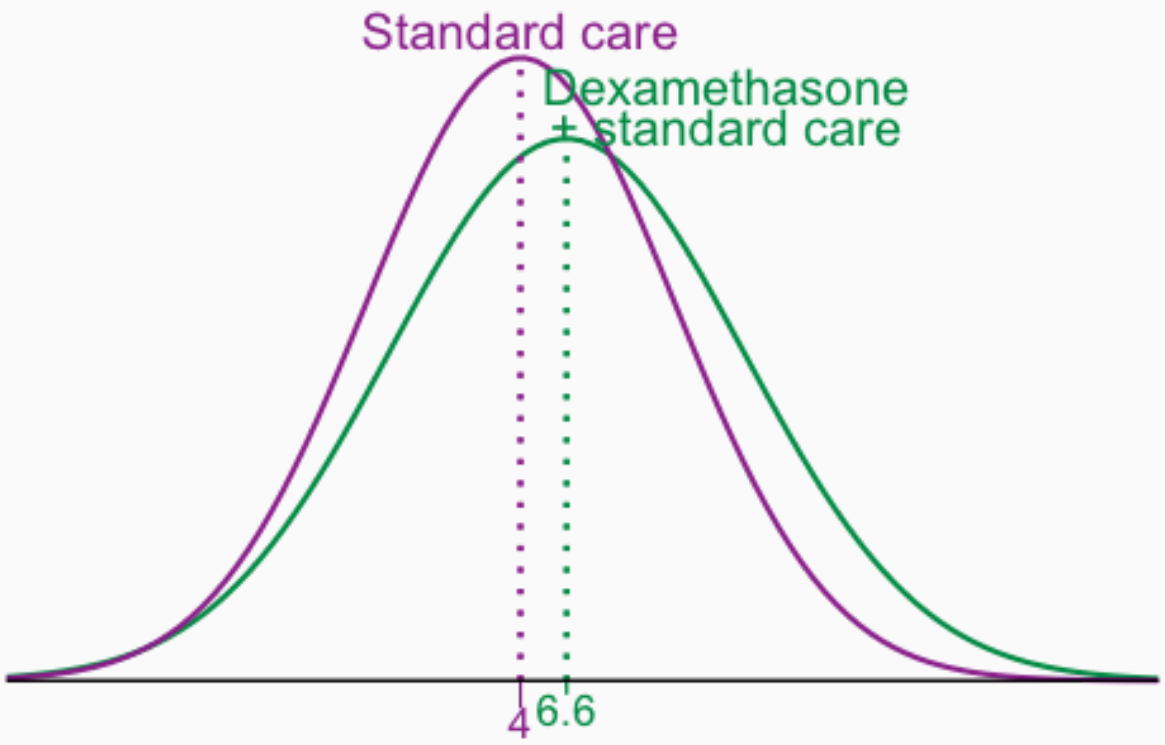
$$n_\mathrm{c} = 148, \bar{x}_\mathrm{c} = 4.0, s_\mathrm{c} = 8.7$$

$$\mu_\mathrm{c} - \mu_\mathrm{i} = 0 \quad \leftarrow$$
$$\bar{x}_\mathrm{c} - \bar{x}_\mathrm{i} = 6.6 - 4.0 = 2.6$$

$$2.6 \neq 0$$

Can we stop here?



Exercise #5

? Even if the null hypothesis were true, the difference between the means might not be exactly zero because of

a) the null hypothesis being formulated imprecisely
b) individual differences
c) measurement errors
d) if the null hypothesis is true, the difference is always zero

Testing hypotheses

📌 Dexamethasone plus standard care is **as effective as** standard care

$$n_\mathrm{i} = 151, \bar{x}_\mathrm{i} = 6.6, s_\mathrm{i} = 10.0$$
$$n_\mathrm{c} = 148, \bar{x}_\mathrm{c} = 4.0, s_\mathrm{c} = 8.7$$

$$\mu_\mathrm{c} - \mu_\mathrm{i} = 0 \quad \leftarrow$$
$$\bar{x}_\mathrm{c} - \bar{x}_\mathrm{i} = 6.6 - 4.0 = 2.6$$

❓ What is the probability of observing a difference of 2.6 days if $\mu_\mathrm{c} - \mu_\mathrm{i} = 0$?

Let's take a step back

1. The Normal distribution is defined by its mean μ and standard deviation σ and represents a probability distribution
   $\rightarrow$ Area under the curve $\equiv$ probability $\mathcal{P}$
2. Sampling distributions (including that of the difference between means) follow a Normal distribution according to the Central Limit Theorem

For the difference of means:

$$\mathcal{N} = (\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}) \text{ with}$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \rightarrow \text{ standard error}$$

Testing hypotheses

📌 Dexamethasone plus standard care is **as effective as** standard care

$$n_{\mathrm{i}} = 151, \bar{x}_{\mathrm{i}} = 6.6, s_{\mathrm{i}} = 10.0$$
$$n_{\mathrm{c}} = 148, \bar{x}_{\mathrm{c}} = 4.0, s_{\mathrm{c}} = 8.7$$

$$\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0 \quad \leftarrow$$
$$\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}} = 6.6 - 4.0 = 2.6$$

$$\mathcal{N} = (\mu_{\mathrm{c}} - \mu_{\mathrm{i}}, \frac{\sigma_c^2}{n_c} + \frac{\sigma_i^2}{n_i}) \rightarrow \mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$$
$$\hat{\mathrm{SE}}^{(*)} = \sqrt{\frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}} + \frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}}} = 1.08$$

$^{(*)}$ In reality, we don't know σ, only s. Therefore, we'll use a Student's t with $\mathrm{df} = (n_{\mathrm{c}} + n_{\mathrm{i}} - 2)$, which we'll approximate to z

Testing hypotheses

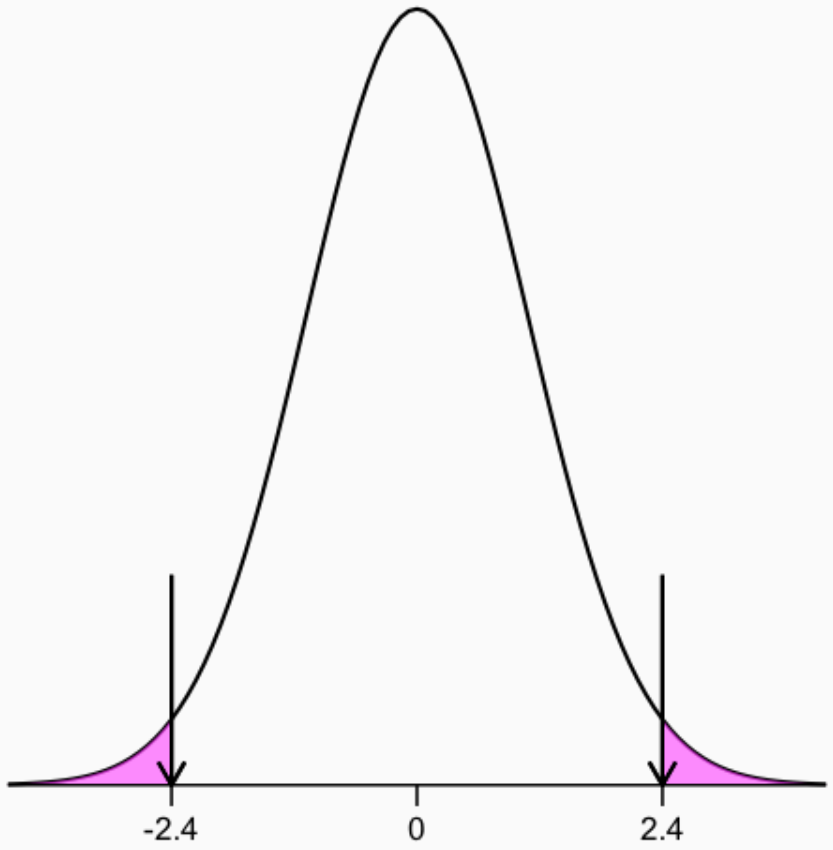
📌 Dexamethasone plus standard care is **as effective as** standard care

$$\mu_c - \mu_i = 0$$
$$\hat{SE} = 1.08$$

$$\bar{x}_c - \bar{x}_i = 6.6 - 4.0 = 2.6$$

❓ What is the probability of observing a difference of 2.6 days if $\mu_c - \mu_i = 0$?

$$z = \frac{(\bar{x}_c - \bar{x}_i) - (\mu_c - \mu_i)}{\hat{SE}} = \frac{2.6 - 0}{1.08} = 2.4$$



Testing hypotheses

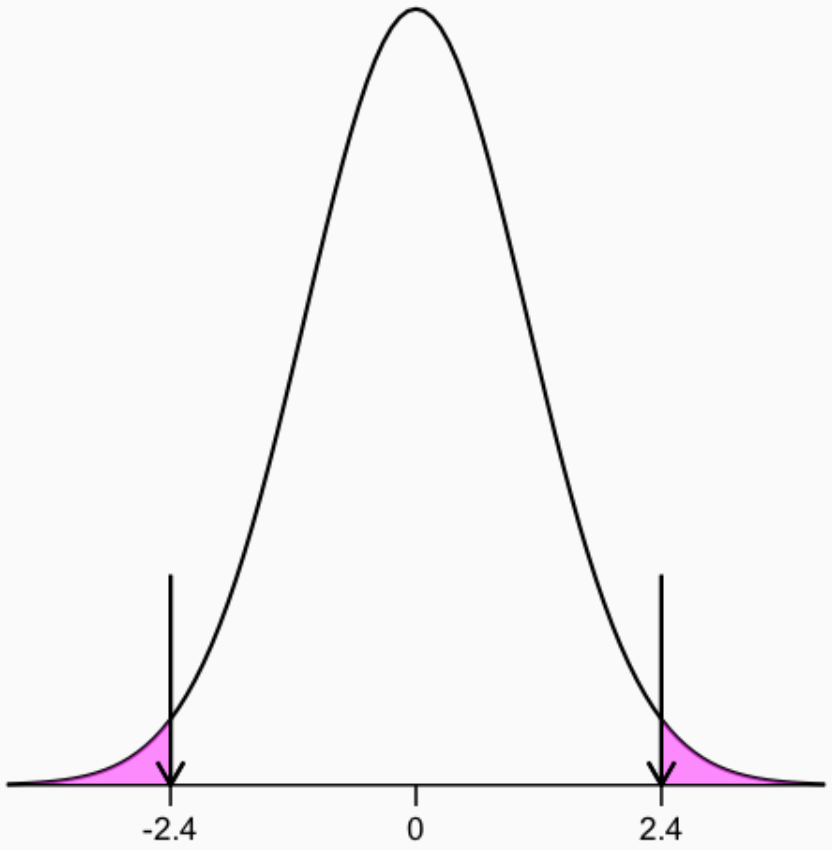
📌 Dexamethasone plus standard care is **as effective as** standard care

$$\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$$
$$\hat{\mathrm{SE}} = 1.08$$

$$\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}} = 6.6 - 4.0 = 2.6$$

❓ What is the probability of observing a difference of 2.6 days if $\mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$?

$$z = 2.4 \rightarrow \mathcal{P} = 2 \times 0.008 = 0.016$$



P-value

🎯 The P-value measures the discrepancy between the data and $\mathcal{H}_0$, and correspond to the probability of observing a result at least as extreme as the one obtained, assuming $\mathcal{H}_0$ is true

$$\text{P-value} = 0.5 \rightarrow 50\% \rightarrow 1 \text{ sample out of } 2$$
$$\text{P-value} = 0.1 \rightarrow 10\% \rightarrow 1 \text{ sample out of } 10$$
$$\text{P-value} = 0.05 \rightarrow 5\% \rightarrow 1 \text{ sample out of } 20$$
$$\text{P-value} = 0.01 \rightarrow 1\% \rightarrow 1 \text{ sample out of } 100$$
$$\text{P-value} = 0.005 \rightarrow 0.5\% \rightarrow 1 \text{ sample out of } 200$$

P-value e statistical significance

🎯 The P-value measures the discrepancy between the data and $\mathcal{H}_0$, and correspond to the probability of observing a result at least as extreme as the one obtained, assuming $\mathcal{H}_0$ is true

If the P-value is smaller than a pre-specified significance level α, we consider the observed difference to be statistically significant

$$\alpha = 0.05 \text{ or } 0.01$$

Why $\alpha = 0.05$?

- *"The value for which P=0.05, or 1 in 20, is 1.96 or nearly 2; it is convenient to take this point as a limit in judging whether a deviation ought to be considered significant or not"*
- *"Either there is something in the treatment, or a coincidence has occurred such as does not occur more than once in twenty trials"*
- *"If one in twenty does not seem high enough odds, we may, if we prefer it, draw the line at one in fifty (the 2 per cent. point), or one in a hundred (the 1 per cent. point)"*

Exercise #6

? When a result in a study is described as "statistically significant", it means that

a) the null hypothesis has been rejected
b) the null hypothesis has **not** been rejected
c) the observed result is probably due to sampling errors
d) the observed result is probably **not** due to sampling errors
e) the p-value is lower than the significance level α
f) the p-value is greater than the significance level α

Exercise #7

? If one **doesn't** reject the null hypothesis it means that...

a) the null hypothesis is true
b) the null hypothesis is false
c) the observations are compatible with the null hypothesis
d) the observations aren't compatible with the null hypothesis
e) it depends on the research question

Hypothesis testing, one step at a time

Hypothesis testing, one step at a time

1. Define a null hypothesis $(\mathcal{H}_0)$

   Dexamethasone plus standard care is **as effective as** standard care

   $\mathcal{H}_0 : \mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0$

Hypothesis testing, one step at a time

1. Define a null hypothesis $(\mathcal{H}_0)$
2. Choose a test statistic that estimates something that, if extreme enough, would lead one to doubt $\mathcal{H}_0$

t-test$^{(*)}$ for differences in mean

$^{(*)}$ Formally, we use the t-test for comparing means rather than the z-test because the population standard deviation is unknown, so we estimate it using the sample standard deviation

Hypothesis testing, one step at a time

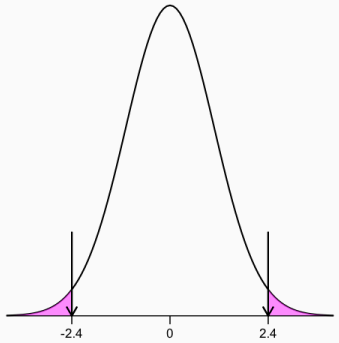
1. Define a null hypothesis $(\mathcal{H}_0)$
2. Choose a test statistic that estimates something that, if extreme enough, would lead one to doubt $\mathcal{H}_0$
3. Generate the sampling distribution of the chosen test statistic, assuming $\mathcal{H}_0$ to be true

$$\mathcal{N} = (\mu_{\mathrm{c}} - \mu_{\mathrm{i}}, \mathrm{SE}), \text{ with } \mu_{\mathrm{c}} - \mu_{\mathrm{i}} = 0 \text{ and } \hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}} + \frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}}}$$

Hypothesis testing, one step at a time

1. Define a null hypothesis $(\mathcal{H}_0)$
2. Choose a test statistic that estimates something that, if extreme enough, would lead one to doubt $\mathcal{H}_0$
3. Generate the sampling distribution of the chosen test statistic, assuming $\mathcal{H}_0$ to be true
4. Check whether the observed statistic lies in the tails of this distribution, and calculate a probability (P-value) for this event

$$\mathcal{P} = 2 \times 0.0082 = 0.0164$$



Hypothesis testing, one step at a time

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3. Generate the sampling distribution of the chosen test statistic, assuming $\mathcal{H}_0$ to be true
4. Check whether the observed statistic lies in the tails of this distribution, and calculate a probability (P-value) for this event
5. Declare the result statistically significant if the P-value is below some critical threshold α

$$\mathcal{P} = 2 \times 0.0082 = 0.0164 < \alpha = 0.05 \quad \rightarrow \quad \text{one rejects } \mathcal{H}_0$$

Share the conclusions

📌 Dexamethasone combined with standard care is not as effective as standard care alone. We observe a statistically significant difference of 2.6 days between the two treatments ($P = 0.016$).

What is the uncertainty associated with this estimate?

Share the conclusions

📌 Dexamethasone combined with standard care is not as effective as standard care alone. We observe a statistically significant difference of 2.6 days between the two treatments (P = 0.016).

What is the uncertainty associated with this estimate?

1. $\hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}} + \frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}}} = 1.08$

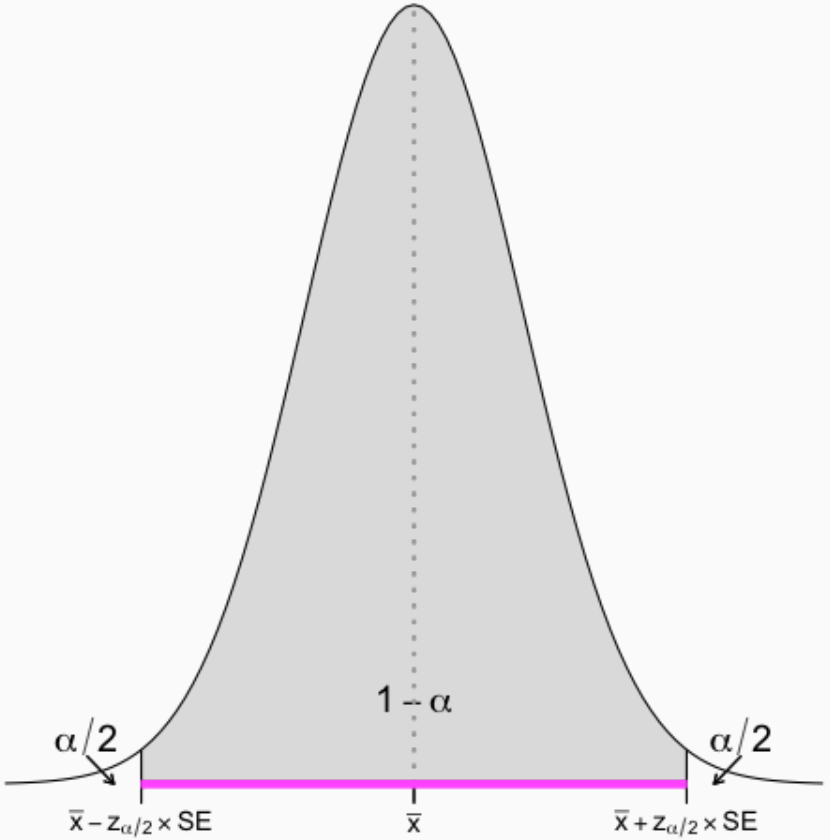
2. $95\%\ \mathrm{ME} = 2 \times \hat{\mathrm{SE}} = 2 \times 1.08 = 2.16$

3. $95\%\ \mathrm{CI} = (\bar{x}_{\mathrm{i}} - \bar{x}_{\mathrm{c}}) - 95\%\ \mathrm{ME}\ ;\ (\bar{x}_{\mathrm{i}} - \bar{x}_{\mathrm{c}}) + 95\%\ \mathrm{ME} =$
$= (2.6 - 2.16\ ;\ 2.6 + 2.16) = (0.44\ ;\ 4.78)$

Hypothesis Testing and Confidence Intervals

🎯 The 95% confidence interval is the set of null hypotheses that would not be rejected at a significance level of $\alpha = 0.05$

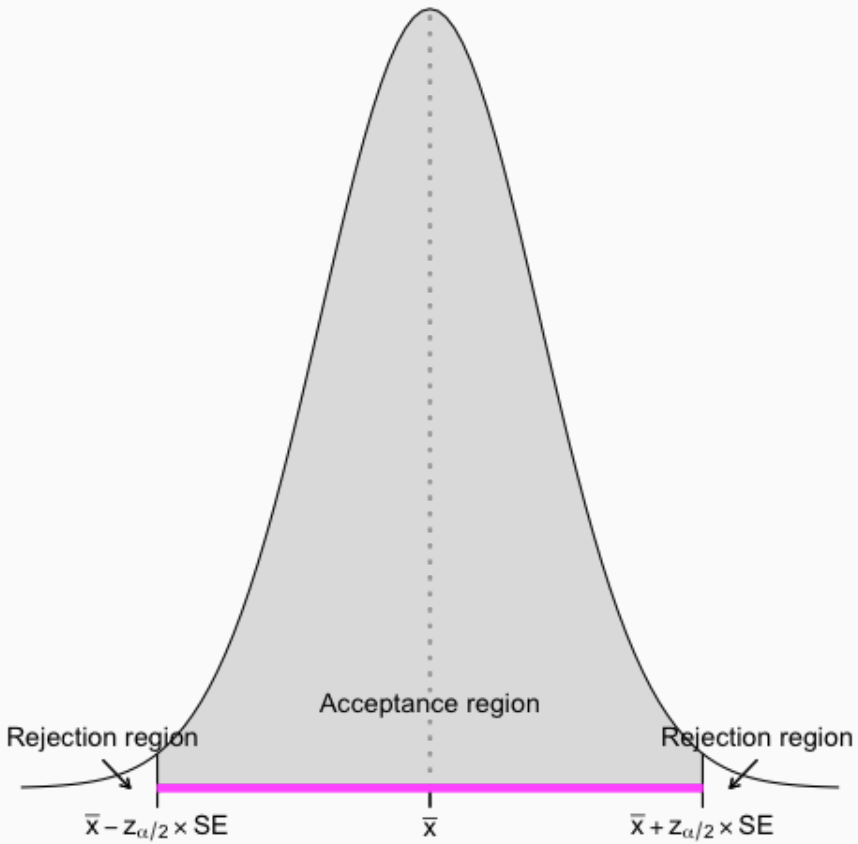
Confidence level	α	$\alpha/2$
95%	5%	2.5%



Hypothesis Testing and Confidence Intervals

🎯 The 95% confidence interval is the set of null hypotheses that would not be rejected at a significance level of $\alpha = 0.05$

In a two-sided test, $\mathrm{P} < 0.05$ if the 95% confidence interval does not include the null hypothesis (usually zero)



Share the conclusions

📌 Dexamethasone combined with standard care is not as effective as standard care alone. We observe a statistically significant difference of 2.6 days (95% CI = 0.44 ; 4.78) between the two treatments (P = 0.016).

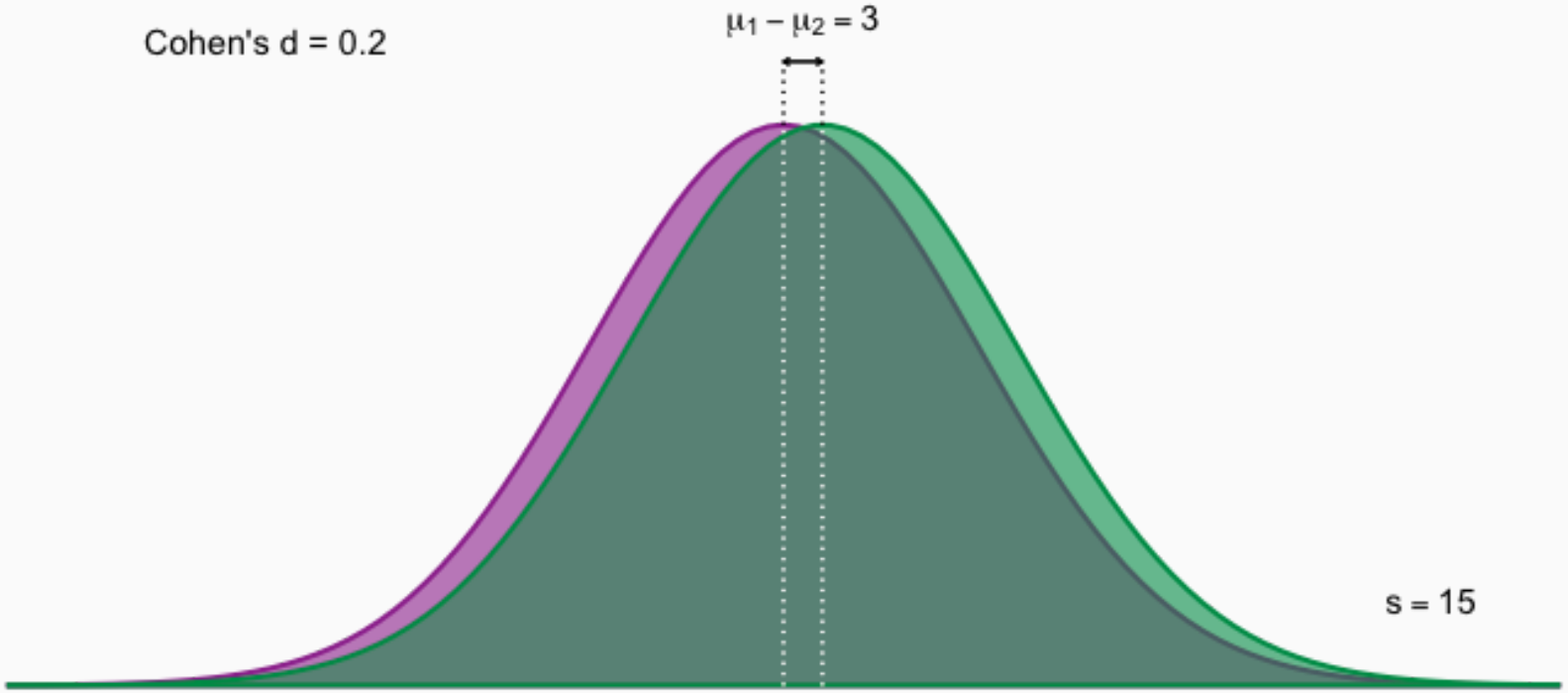
What is the clinical significance of the treatment, considering that the true population difference could be as small as 0.44 days?

Effect size

📌 $\text{Cohen's } d = \left|\frac{\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}}}{s_p}\right|$ con $s_p = \sqrt{\frac{(n_i - 1) \times s_i + (n_c - 1) \times s_c}{(n_i - 1) + (n_c - 1)}}$

Effect size

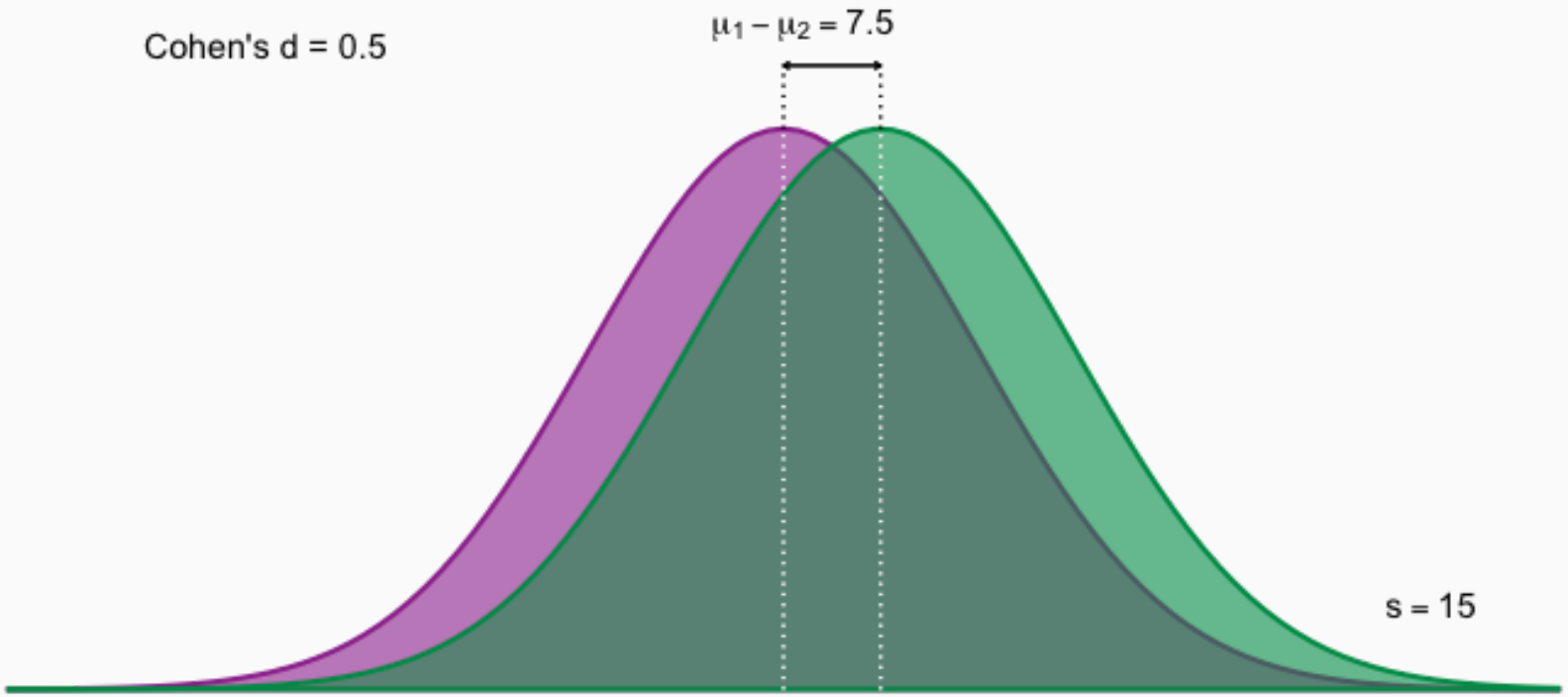
📌 $$\text{Cohen's } d = \left|\frac{\bar{x}_c - \bar{x}_i}{s_p}\right| \quad \text{with} \quad s_p = \sqrt{\frac{(n_i-1)\times s_i + (n_c-1)\times s_c}{(n_i-1)+(n_c-1)}}$$



d	Interpretation
0.2	Small

Effect size

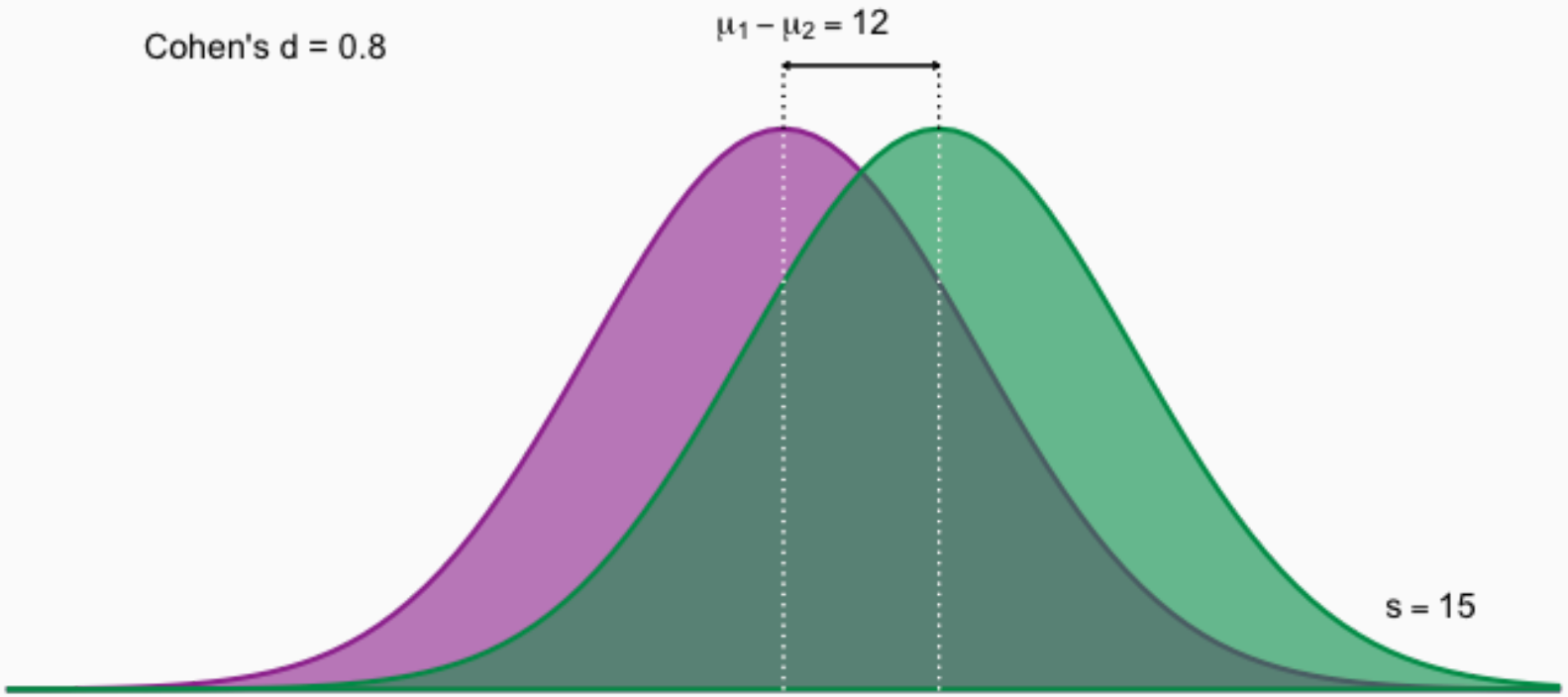
📌 $$\text{Cohen's } d = \left|\frac{\bar{x}_c - \bar{x}_i}{s_p}\right| \quad \text{with} \quad s_p = \sqrt{\frac{(n_i-1)\times s_i + (n_c-1)\times s_c}{(n_i-1)+(n_c-1)}}$$



d	Interpretation
0.2	Small
0.5	Medium

Effect size

📌 $$\text{Cohen's } d = \left|\frac{\bar{x}_c - \bar{x}_i}{s_p}\right| \quad \text{with} \quad s_p = \sqrt{\frac{(n_i-1)\times s_i + (n_c-1)\times s_c}{(n_i-1)+(n_c-1)}}$$



d	Interpretation
0.2	Small
0.5	Medium
0.8	Large

Effect size

📌 Dexamethasone combined with standard care is not as effective as standard care alone. We observe a statistically significant difference of 2.6 days (95% CI = 0.44 ; 4.78) between the two treatments (P = 0.016).

$$\text{Cohen's } d = \left|\frac{\bar{x}_{\mathrm{c}} - \bar{x}_{\mathrm{i}}}{s_p}\right| \quad \text{with} \quad s_p = \sqrt{\frac{(n_i - 1) \times s_i + (n_c - 1) \times s_c}{(n_i - 1) + (n_c - 1)}}$$

$$s_p = \sqrt{\frac{(151-1) \times 10 + (148-1) \times 8.7}{(151-1) + (148-1)}} = 3$$

$$d = \frac{2.6}{3} = 0.85 \quad \rightarrow \quad \text{Large}$$

Share the conclusions

📌 Dexamethasone combined with standard care is not as effective as standard care alone. We observe a statistically significant difference of 2.6 days (95% CI = 0.44 ; 4.78) between the two treatments (P = 0.016). This suggests that the combined treatment has a large effect on the number of ventilator-free days (Cohen's d = 0.85).

Share (and interpret) the conclusions

- **Hypothesis testing:** is the procedure used to evaluate whether the observed data provide sufficient evidence to support a hypothesis
- **Confidence interval:** quantifies the uncertainty around a statistic and estimates a range of plausible values where the true population value is likely to lie
- **Effect size:** The magnitude of the study results, indicating whether the findings are large enough to be meaningful in the real world

Share (and interpret) the conclusions

Scenario 1

- Hypothesis testing: P value $< \alpha$
- Confidence interval: very narrow
- Effect size: medium or large

We have three pieces of evidence supporting the significance of the result

Share (and interpret) the conclusions

Scenario 2

- Hypothesis testing: P value $< \alpha$
- Confidence interval: very narrow
- Effect size: very small or small

We have two pieces of evidence supporting the significance of the result, but the clinical significance is negligible

Share (and interpret) the conclusions

Scenario 3

- Hypothesis testing: P value $> \alpha$
- Confidence interval: very large
- Effect size: very small or small

We have three pieces of evidence supporting the lack of significance of the result

Share (and interpret) the conclusions

Scenario 4

- Hypothesis testing: P value $> \alpha$
- Confidence interval: very large
- Effect size: large

Likely, our sample is too small to determine
with confidence whether to reject the null hypothesis

t-test

📌 For the difference between two means

$$\mathcal{N} = (\mu_{\mathrm{c}} - \mu_{\mathrm{i}}, \frac{\sigma_c^2}{n_c} + \frac{\sigma_i^2}{n_i})$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{s_{\mathrm{c}}^2}{n_{\mathrm{c}}} + \frac{s_{\mathrm{i}}^2}{n_{\mathrm{i}}}}$$

📌 For the difference between two proportions

$$\mathcal{N} = (\pi_{\mathrm{c}} - \pi_{\mathrm{i}}, \frac{\pi_{\mathrm{c}} \times (1 - \pi_{\mathrm{c}})}{n_{\mathrm{c}}} + \frac{\pi_{\mathrm{i}} \times (1 - \pi_{\mathrm{i}})}{n_{\mathrm{i}}})$$

$$\hat{\mathrm{SE}} = \sqrt{\frac{\bar{p}_{\mathrm{c}} \times (1 - \bar{p}_{\mathrm{c}})}{n_{\mathrm{c}}} + \frac{\bar{p}_{\mathrm{i}} \times (1 - \bar{p}_{\mathrm{i}})}{n_{\mathrm{i}}}}$$

Exercise #8

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

> From April 1991 through December 20, 1993, the cutoff date for the first interim analysis of efficacy, 477 pregnant women were enrolled; during the study period, 409 gave birth to 415 live-born infants. HIV-infection status was known for 363 births (180 in the zidovudine group and 183 in the placebo group). Thirteen infants in the zidovudine group and 40 in the placebo group were HIV-infected.

$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad \bar{p}_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad \bar{p}_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

$$\mathcal{N} = (\pi_{\mathrm{c}} - \pi_{\mathrm{i}}, \frac{\pi_{\mathrm{c}} \times (1-\pi_{\mathrm{c}})}{n_{\mathrm{c}}} + \frac{\pi_{\mathrm{i}} \times (1-\pi_{\mathrm{i}})}{n_{\mathrm{i}}})$$

Connor, E.M. *et al.*, *Reduction of Maternal-Infant Transmission of Human Immunodeficiency Virus Type 1 with Zidovudine Treatment*, NEJM, 1994

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

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$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad \bar{p}_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

Let's use another test to compare differences in proportion!

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

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$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad \bar{p}_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

1. Define a null hypothesis $(\mathcal{H}_0)$
   The rate of mother-to-infant HIV transmission is **independent** of zidovudine use

Pearson's χ^2 test

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$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad \bar{p}_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
$$n_{\mathrm{c}} = 183, \quad m_{\mathrm{c}} = 40, \quad \bar{p}_{\mathrm{c}} = \frac{m_{\mathrm{c}}}{n_{\mathrm{c}}} = \frac{40}{183} = 0.22$$

2. Choose a test statistic that estimates something that, if extreme enough, would lead one to doubt $\mathcal{H}_0$

   $\rightarrow$ Pearson's χ^2 test for categorical data

Pearson's χ^2 test

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3. Generate the sampling distribution of the chosen test statistic, assuming $\mathcal{H}_0$ to be true

Pearson's χ^2 test

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3. Generate the sampling distribution of the chosen test statistic, assuming $\mathcal{H}_0$ to be true

? Let's fill this contingency table

Treatment/Infected	Yes	No	Total
Zidovudine			
Placebo			
Total			

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

$$n_{\mathrm{i}} = 180, \quad m_{\mathrm{i}} = 13, \quad \bar{p}_{\mathrm{i}} = \frac{m_{\mathrm{i}}}{n_{\mathrm{i}}} = \frac{13}{180} = 0.07$$
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3. Generate the sampling distribution assuming $\mathcal{H}_0$ to be true

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Pearson's χ^2 test

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$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Expected values

Treatment/Infected	Yes	No	Total
Zidovudine	180*0.146		180
Placebo	183*0.146		183
Total	53	310	363

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?

$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Expected values

Treatment/Infected	Yes	No	Total
Zidovudine	26.28		180
Placebo	26.72		183
Total	53	310	363

Pearson's χ^2 test

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$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Expected values

Treatment/Infected	Yes	No	Total
Zidovudine	26.28	153.72	180
Placebo	26.72	156.28	183
Total	53	310	363

Pearson's χ^2 test

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$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

Observed values

Treatment/Infected	Yes	No	Total
Zidovudine	13	167	180
Placebo	40	143	183
Total	53	310	363

Expected values

Treatment/Infected	Yes	No	Total
Zidovudine	26.28	153.72	180
Placebo	26.72	156.28	183
Total	53	310	363

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected} = \frac{(13-26.28)^2}{26.28} + \frac{(167-153.72)^2}{153.72} + \frac{(40-26.72)^2}{26.72} + \frac{(143-156.28)^2}{156.28} = 15.57$$

Pearson's χ^2 test

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$$\Pi = \frac{tot_{\text{infected}}}{total} = \frac{53}{363} = 0.146$$

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Treatment/Infected	Yes	No	Total
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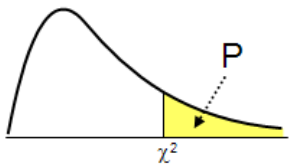
$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected} = \frac{(13-26.28)^2}{26.28} + \frac{(167-153.72)^2}{153.72} + \frac{(40-26.72)^2}{26.72} + \frac{(143-156.28)^2}{156.28} = 15.57$$

$$\text{df} = (n_{\text{rows}} - 1) \times (n_{\text{columns}} - 1) = 1$$

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?



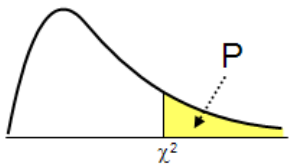
	P										
DF	**0.995**	**0.975**	**0.20**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.002**	**0.001**
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458

4. Check whether the observed statistic lies in the tails of this distribution, and calculate a probability (P-value) for this event

$$\chi^2 = 15.57 \qquad \mathrm{df} = 1 \qquad \rightarrow \qquad \mathrm{P} < 0.001 = 7.9 \times 10^{-5}$$

Pearson's χ^2 test

? Is zidovudine **not** as effective as placebo in reducing mother-to-infant HIV transmission?



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6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458

4. Declare the result statistically significant if the P-value is below some critical threshold α

$$\chi^2 = 15.57 \qquad \mathrm{df} = 1 \qquad \rightarrow \qquad \mathrm{P} < 0.001 = 7.9 \times 10^{-5} < \alpha = 0.05 \rightarrow \text{reject } \mathcal{H}_0$$

Pearson's χ^2 test

📌 Does education level influences physical activity frequency?

Observed values

	No Exercise	Sporadic Exercise	Regular Exercise	Total
Primary education				
Secondary education				
Bachelor/Master				
Doctorate				
Total				

Pearson's χ^2 test

📌 Does education level influences physical activity frequency?

Expected values

	No Exercise	Sporadic Exercise	Regular Exercise	Total
Primary education	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_1}{\text{Total}}$	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_2}{\text{Total}}$	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_3}{\text{Total}}$	ΣRow_1
Secondary education	$\frac{\Sigma\text{Row}_2 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_2
Bachelor/Master	$\frac{\Sigma\text{Row}_3 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_3
Doctorate	$\frac{\Sigma\text{Row}_4 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_4
Total	ΣColumn_1	ΣColumn_2	ΣColumn_3	Total

❓ $\text{df} = \ ?$

Pearson's χ^2 test

📌 Does education level influences physical activity frequency?

Expected values

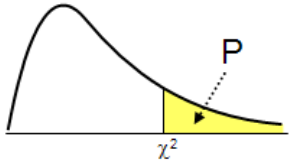
	No Exercise	Sporadic Exercise	Regular Exercise	Total
Primary education	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_1}{\text{Total}}$	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_2}{\text{Total}}$	$\frac{\Sigma\text{Row}_1 \times \Sigma\text{Column}_3}{\text{Total}}$	ΣRow_1
Secondary education	$\frac{\Sigma\text{Row}_2 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_2
Bachelor/Master	$\frac{\Sigma\text{Row}_3 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_3
Doctorate	$\frac{\Sigma\text{Row}_4 \times \Sigma\text{Column}_1}{\text{Total}}$	...	...	ΣRow_4
Total	ΣColumn_1	ΣColumn_2	ΣColumn_3	Total

❓ $\text{df} = (n_{\text{row}} - 1) \times (n_{\text{column}} - 1) = (4 - 1) \times (3 - 1) = 3 \times 2 = 6$

Exercise #9

? Does the area of practice influences drinking habits of Italian healthcare workers?

Out of 279, 230, and 130 healthcare professionals working in medicine, surgery, and other wards (*e.g.*, laboratories and other hospital services), 122, 107, and 51 were non-drinkers, respectively.



	P										
DF	**0.995**	**0.975**	**0.20**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.002**	**0.001**
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Albano, L. *et al.*, *Alcohol consumption in a sample of Italian healthcare workers: A cross-sectional study*, Archives of Environmental & Occupational Health, 2020

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		
False		

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		False positive
False	False negative	

Errors in decision making

Suspect is	Absolved	Convicted
Innocent		One convicts an innocent
Guilty	One absolve an offender	

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		Type I error
False	Type II error	

Exercise #10

? There was a shepherd boy who repeatedly cried wolf when there was no wolf. Yet, each time, villagers went to help him. Then, the wolf arrived, but, when the boy cried wolf, no villager helped.

First we see an example of a...

a) Type I error, then Type II error
b) Type II error, then Type I error
c) Null error, then alternative error
d) Alternative error, then null error

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		Type I error (α)
False	Type II error	

$p < \alpha \rightarrow \text{reject } \mathcal{H}_0$
$p \geq \alpha \rightarrow \text{does not reject } \mathcal{H}_0$

$\alpha = 0.05 \rightarrow 5\%$ chance of rejecting $\mathcal{H}_0$ when is true

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		Type I error (α)
False	Type II error (β)	

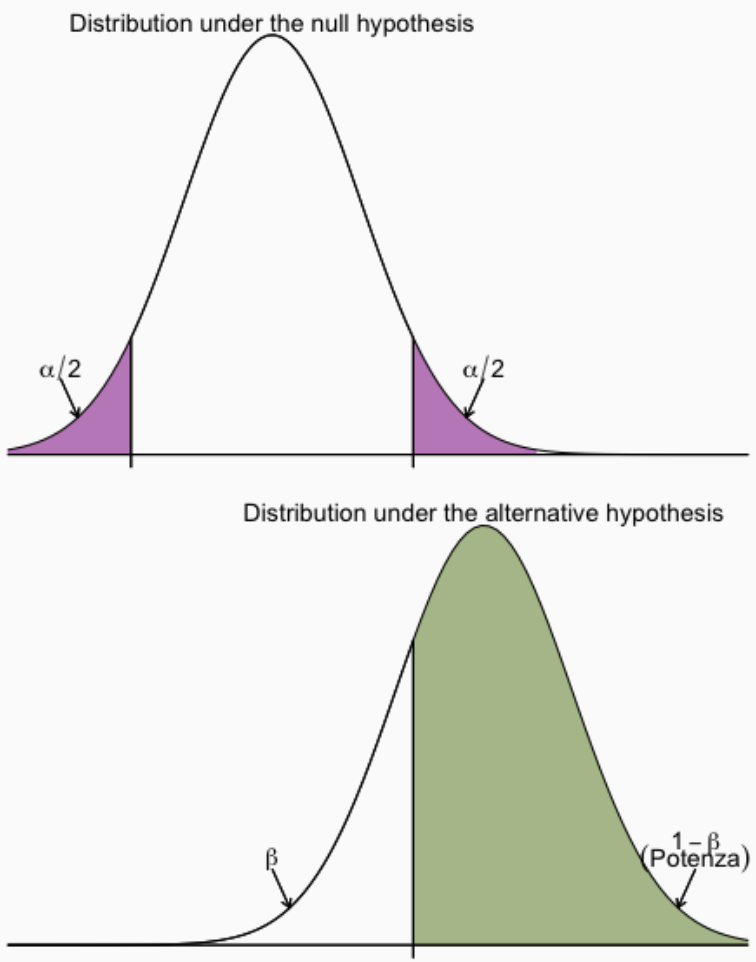
Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		Type I error (α)
False	Type II error (β)	$1-\beta$ (Statistical power)

$$\alpha = 0.05$$
$$1-\beta = 0.8$$

The power of a study

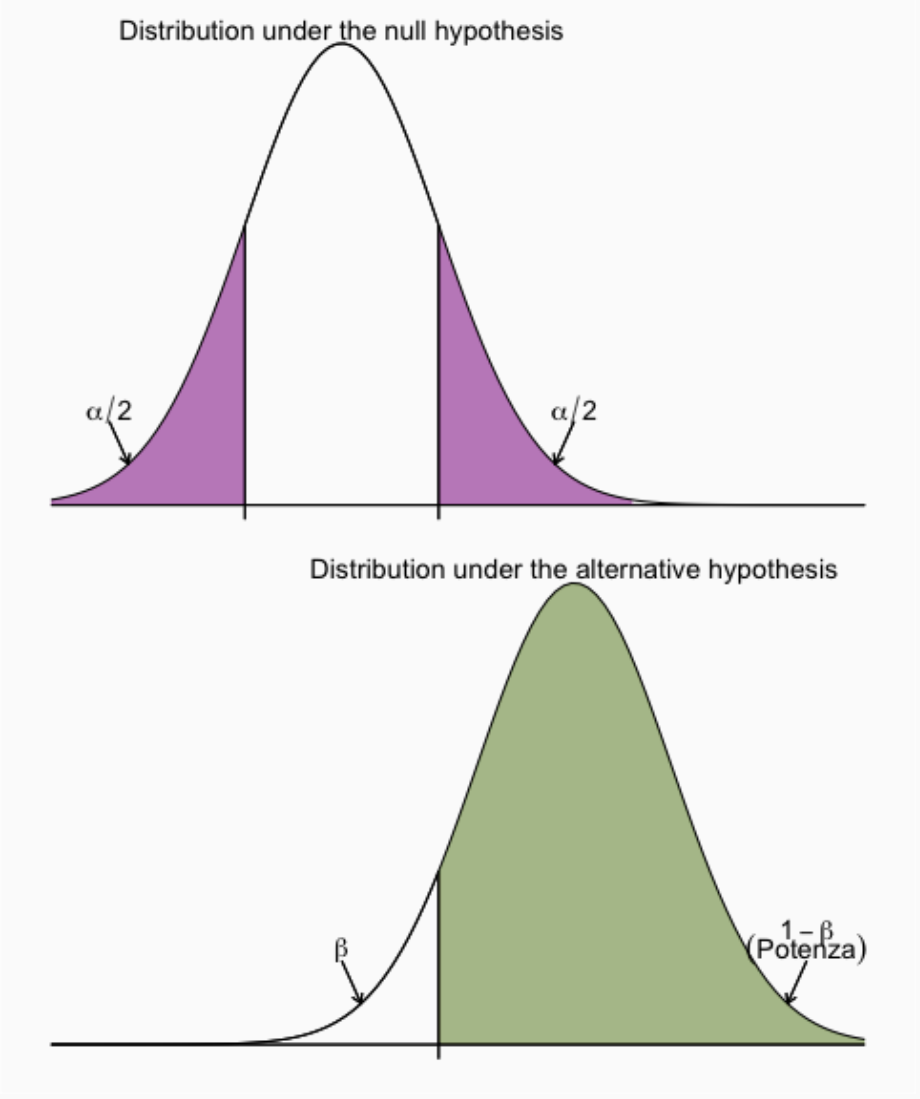
$\mathcal{H}_0$ is	Not rejected	Rejected
True		α
False	β	$1-\beta$ Power



The power of a study

🎯 The power is increased by:

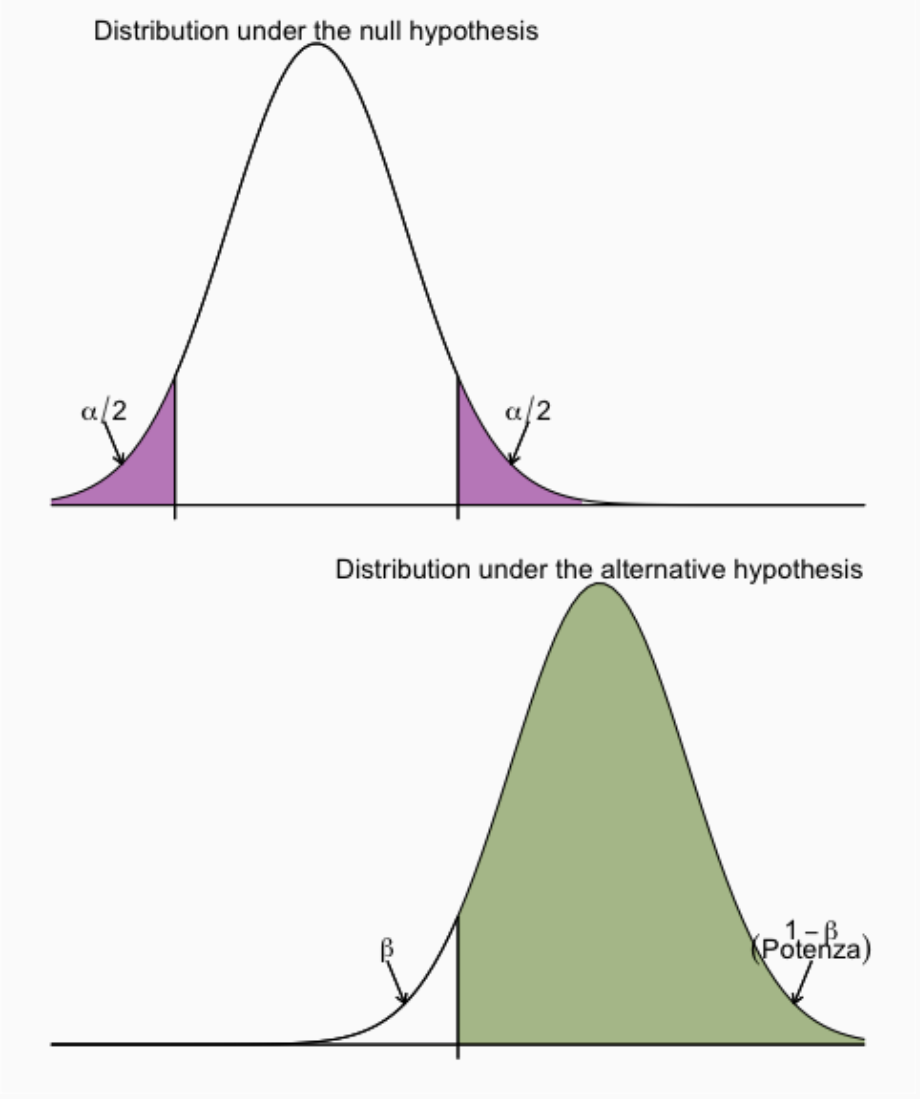
- larger α



The power of a study

🎯 The power is increased by:

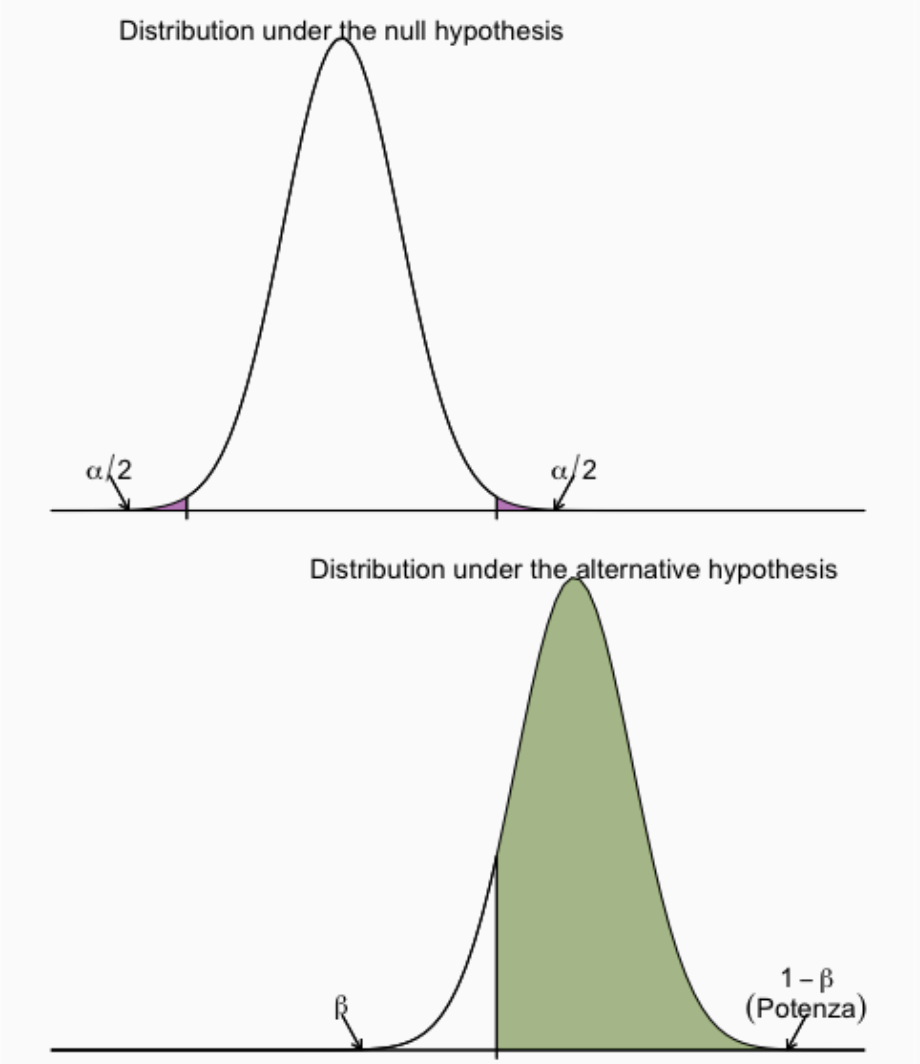
- larger α
- larger $\mu_c - \mu_i$ or $\pi_c - \pi_i$



The power of a study

🎯 The power is increased by:

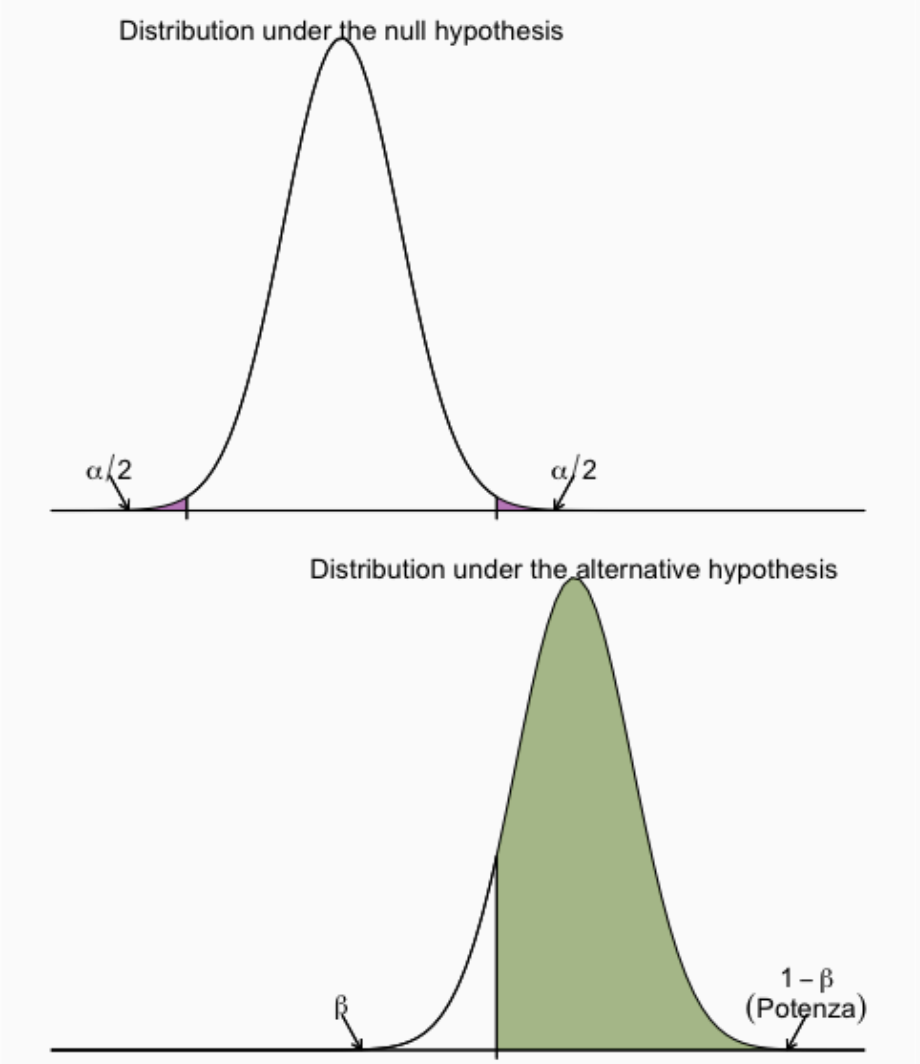
- larger α
- larger $\mu_c - \mu_i$ or $\pi_c - \pi_i$
- smaller σ^2



The power of a study

🎯 The power is increased by:

- larger α
- larger $\mu_c - \mu_i$ or $\pi_c - \pi_i$
- smaller σ^2
- larger sample size n



Exercise #11

? Which of these factors can be modified to increase the statistical power of a study?)

a) the level of significance α
b) the difference $\mu_{\mathrm{c}} - \mu_{\mathrm{i}}$ / $\pi_{\mathrm{c}} - \pi_{\mathrm{i}}$
c) the samples' σ^2
d) the samples' size n

What's the deal with power?

🎯 Power $= 1 - \beta \rightarrow$ chance of rejecting $\mathcal{H}_0$ when is false

- larger $\mu_{\mathrm{c}} - \mu_{\mathrm{i}}$ or $\pi_{\mathrm{c}} - \pi_{\mathrm{i}} \longrightarrow$ larger effect size
- larger sample size n

Observation #1: because any intervention likely has some real effect, by collecting a sufficiently large amount of data, one can always find a statistically significant result.
The effect size, however, will be extremely tiny and probably clinically irrelevant

What’s the deal with power?

🎯 Power $= 1 - \beta \rightarrow$ chance of rejecting $\mathcal{H}_0$ when is false

- larger $\mu_{\mathrm{c}} - \mu_{\mathrm{i}}$ or $\pi_{\mathrm{c}} - \pi_{\mathrm{i}} \longrightarrow$ larger effect size
- larger sample size n

Observation #2: if one doesn't collect a sufficiently large amount of data, they can only detect very large effect sizes

What’s the deal with power?

🎯 Power $= 1 - \beta \rightarrow$ chance of rejecting $\mathcal{H}_0$ when is false

- larger $\mu_{\mathrm{c}} - \mu_{\mathrm{i}}$ or $\pi_{\mathrm{c}} - \pi_{\mathrm{i}} \longrightarrow$ larger effect size
- larger sample size n

Observation #2: if one doesn't collect a sufficiently large amount of data, they can only detect very large effect sizes

$\longrightarrow$ they may conclude that "*there was no statistically significant difference between groups*”, when actually there was insufficient data to detect any but the largest difference: **the study is underpowered**.

What’s the deal with power?

Scenario 4

- Hypothesis testing: P value $> \alpha$
- Confidence interval: very large
- Effect size: large

Likely, our sample is too small to determine
with confidence whether to reject the null hypothesis

Our study is underpowered

What’s the deal with power?

🎯 Power $= 1 - \beta \rightarrow$ chance of rejecting $\mathcal{H}_0$ when is false

- larger $\mu_{\mathrm{c}} - \mu_{\mathrm{i}}$ or $\pi_{\mathrm{c}} - \pi_{\mathrm{i}} \longrightarrow$ larger effect size
- larger sample size n

Observation #2: if one doesn't collect a sufficiently large amount of data, they can only detect very large effect sizes

$\longrightarrow$ they may conclude that "*there was a very large difference between groups which was statistically significant*": **the effect size is inflated**

Type M error

Making multiple comparisons

📌 A research group showed to a single subject (*) a series of pics of humans expressing different emotions while carrying out brain imaging (fMRI). They identified 16 out of brain regions showing a 130,000 significant response at $P < 0.001$.

Making multiple comparisons

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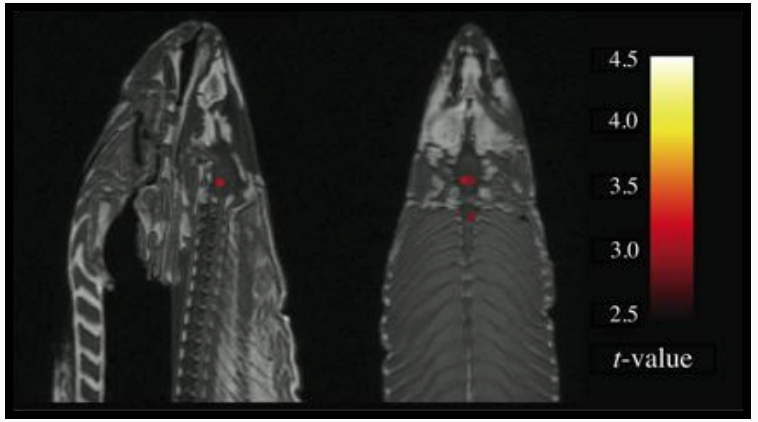

(*) Atlantic salmon, '*not alive at the time of scanning*'

Bennett, C. M., Miller M.B., and Wolford G.L.,. *Neural correlates of interspecies perspective taking in the post-mortem Atlantic Salmon: An argument for multiple comparisons correction.* Neuroimage 47.Suppl 1 (2009) doi:10.1016/S1053-8119(09)71202-9

Errors in decision making

$\mathcal{H}_0$ is	Not rejected	Rejected
True		Type I error (α)
False	Type II error	

$\alpha = 0.05 \rightarrow 5\%$ chance of rejecting $\mathcal{H}_0$ when is true
$1 - \alpha = 0.95 \rightarrow 95\%$ chance of not rejecting $\mathcal{H}_0$ when is true

Making multiple comparisons

$\alpha = 0.05 \rightarrow 5\%$ chance of rejecting $\mathcal{H}_0$ when is true
$1 - \alpha = 0.95 \rightarrow 95\%$ chance of not rejecting $\mathcal{H}_0$ when is true

Probability of finding at least a statistically significant result

- with 1 test $\longrightarrow 1 - 0.95 = 0.05 = 5\%$

Making multiple comparisons

$\alpha = 0.05 \rightarrow 5\%$ chance of rejecting $\mathcal{H}_0$ when is true
$1 - \alpha = 0.95 \rightarrow 95\%$ chance of not rejecting $\mathcal{H}_0$ when is true

Probability of finding at least a statistically significant result

- with 1 test $\longrightarrow 1 - 0.95 = 0.05 = 5\%$
- with 2 tests $\longrightarrow 1 - 0.95 \times 0.95 = 1 - 0.95^2 = 0.0975 \approx 10\%$

Making multiple comparisons

$\alpha = 0.05 \rightarrow 5\%$ chance of rejecting $\mathcal{H}_0$ when is true
$1 - \alpha = 0.95 \rightarrow 95\%$ chance of not rejecting $\mathcal{H}_0$ when is true

Probability of finding at least a statistically significant result

- with 1 test $\longrightarrow 1 - 0.95 = 0.05 = 5\%$
- with 2 tests $\longrightarrow 1 - 0.95 \times 0.95 = 1 - 0.95^2 = 0.0975 \approx 10\%$
- with 3 tests $\longrightarrow 1 - 0.95^3 = 0.142625 \approx 14\%$

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- with 130,000 tests $\longrightarrow 1 - 0.95^{130,000} = 1 = 100\%$

$$1 - (1 - \alpha)^{N_{\text{tests}}}$$

Correcting for multiple comparisons

🎯 Using a smaller significance level α to determine statistical significance

Bonferroni correction: $\frac{\alpha}{N_{\text{tests}}} \longrightarrow \frac{0.05}{N_{\text{tests}}}$

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- with 1 test $\rightarrow 1-(1-\frac{0.05}{1})^1 = 1-(1-0.05)^1 = 1-0.95^1 = 0.05$
- with 2 tests $\rightarrow 1-(1-\frac{0.05}{2})^2 = 1-(1-0.025)^2 = 1-0.975^2 = 0.049$
- with 3 tests $\rightarrow 1-(1-\frac{0.05}{3})^3 = 1-(1-0.0167)^3 = 1-0.983^3 = 0.050$
- with 10 tests $\rightarrow 1-(1-\frac{0.05}{10})^{10} = 1-(1-0.005)^{10} = 1-0.995^{10} = 0.049$
- with 130,000 tests $\rightarrow 1-(1-\frac{0.05}{130{,}000})^{130{,}000} = 1-(1-4\times 10^{-7})^{130{,}000} =$

$$= 1-0.999^{130{,}000} = 0.049$$

Correcting for multiple comparisons

🎯 Controlling the **expected** proportion of false discoveries

False discovery rate (FDR, Benjamini–Hochberg procedure):

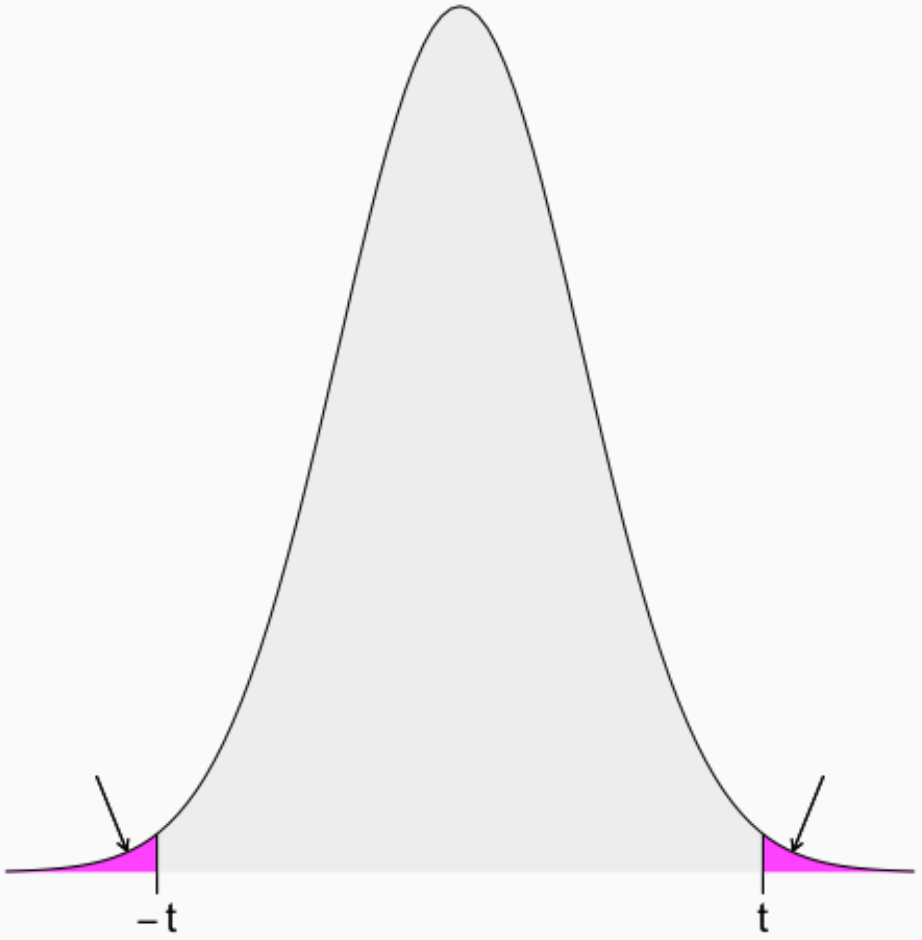
1. Sort test results from the smallest to the largest P-value
2. For a given α, find the largest k such that $\mathcal{P}(k) \leq \frac{k}{m}\alpha$
3. Reject the null hypothesis for $i = 1, \ldots, k$

One- and two-tailed tests

🎯 $\mathcal{H}_1$: $\mu_\mathrm{c} - \mu_\mathrm{i} \neq 0$

$\mathcal{H}_0$: $\mu_\mathrm{c} - \mu_\mathrm{i} = 0$

$\longrightarrow$ two-tailed test



One- and two-tailed tests

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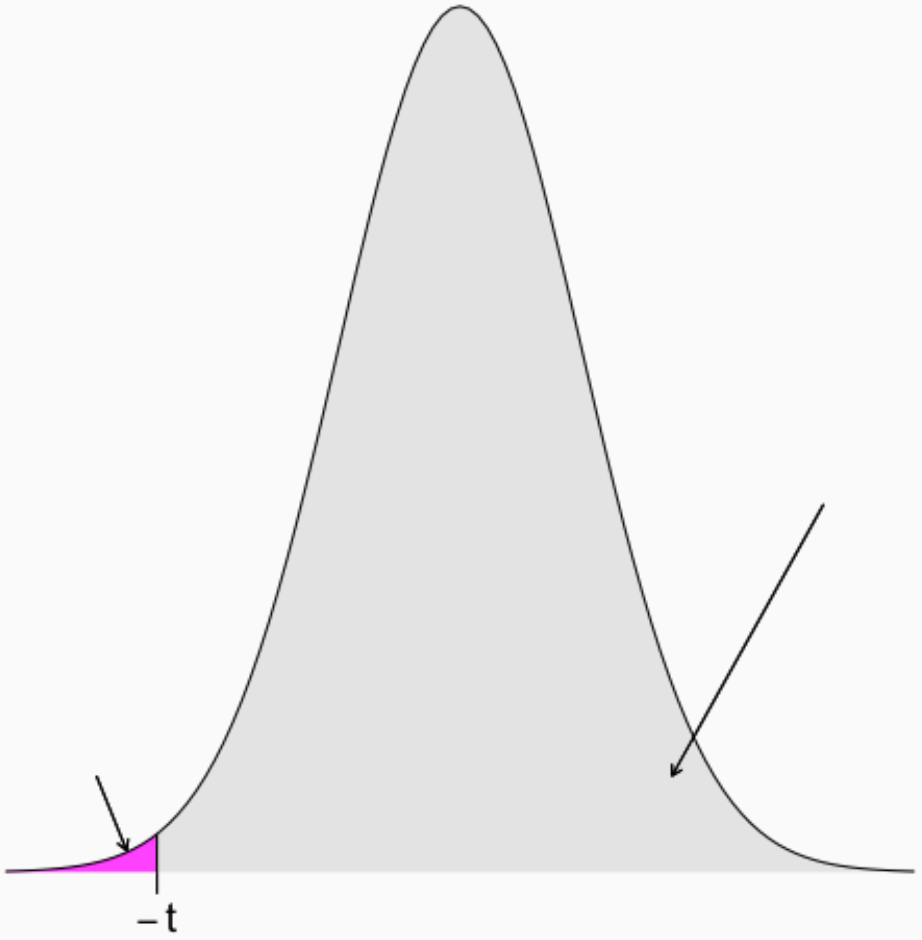
$\mathcal{H}_0$: $\mu_\mathrm{c} - \mu_\mathrm{i} = 0$

$\longrightarrow$ two-tailed test

$\mathcal{H}_1$: $\mu_\mathrm{c} - \mu_\mathrm{i} < 0$
$\mathcal{H}_0$: $\mu_\mathrm{c} - \mu_\mathrm{i} \geq 0$

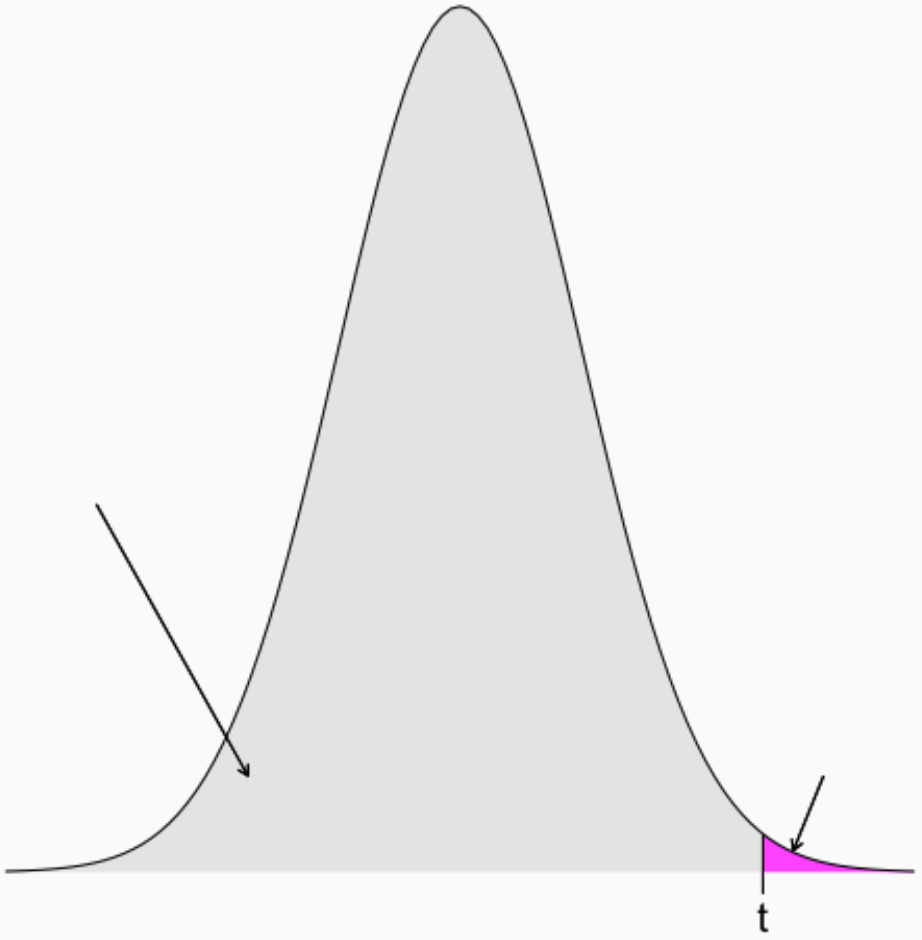
$\longrightarrow$ one-tailed test



One- and two-tailed tests

🎯 $\mathcal{H}_1$: $\mu_c - \mu_i \neq 0$
$\mathcal{H}_0$: $\mu_c - \mu_i = 0$
$\rightarrow$ two-tailed test

$\mathcal{H}_1$: $\mu_c - \mu_i < 0$
$\mathcal{H}_0$: $\mu_c - \mu_i \geq 0$
or
$\mathcal{H}_1$: $\mu_c - \mu_i > 0$
$\mathcal{H}_0$: $\mu_c - \mu_i \leq 0$
$\rightarrow$ one-tailed test

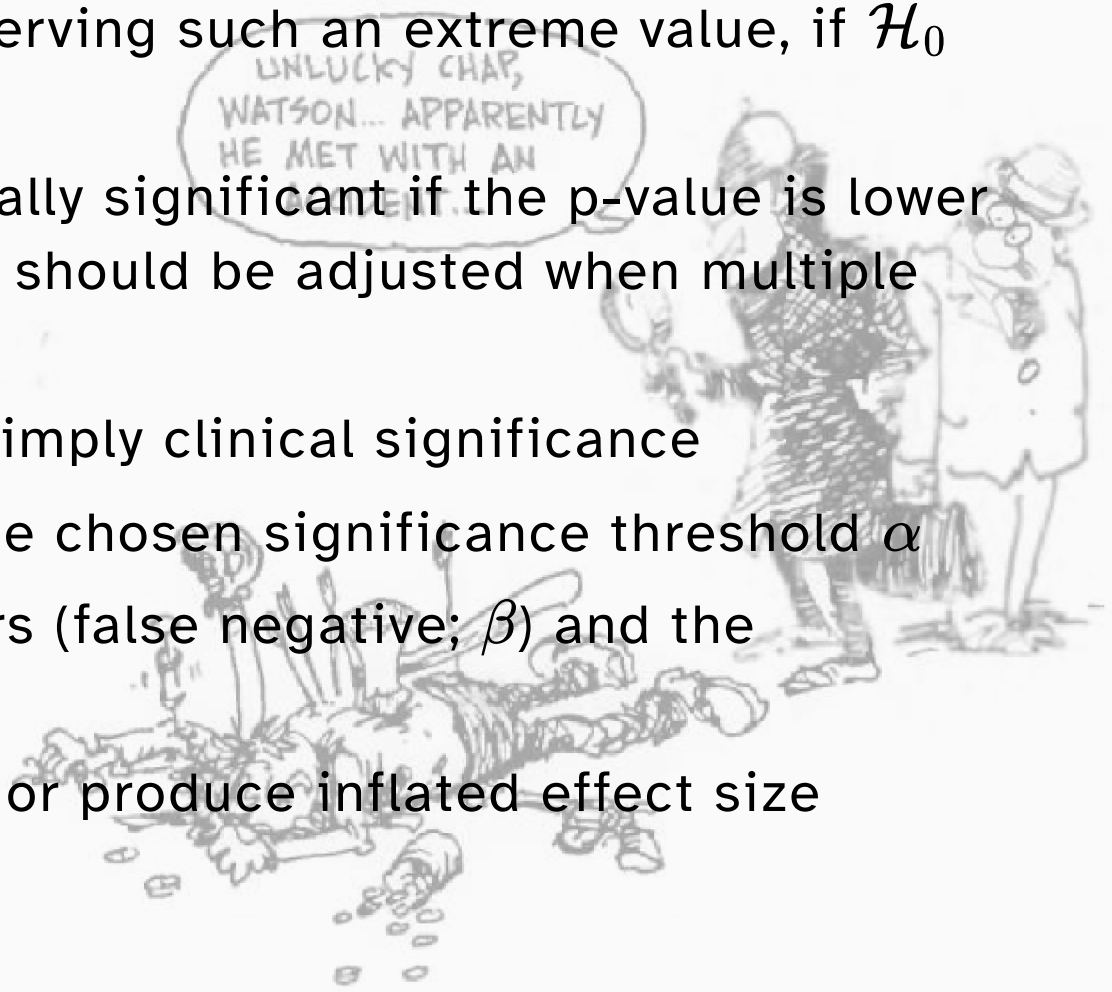


Non-parametric tests

Sample	Data type	$\mathcal{H}_0$	Non-parametric test
Independent	Numerical	$\mu_1 = \mu_2$	Mann-Whitney's test
Paired	Numerical	$\mu_1 = \mu_2$	Wilcoxon's test
Independent	Categorical	$\pi_1 = \pi_2$	Fisher's test
Paired	Categorical	$\pi_1 = \pi_2$	McNemar's test

Summary

- The P-value measures the discrepancy between the data and the null hypothesis $\mathcal{H}_0$, and correspond to the probability of observing such an extreme value, if $\mathcal{H}_0$ was true
- Traditionally, results are considered statistically significant if the p-value is lower than α = 0.05$ or 0.01, but these thresholds should be adjusted when multiple tests are performed.
- Statistical significance does not necessarily imply clinical significance
- Type I errors (or false positive) depend on the chosen significance threshold α
- There is a relationship between Type II errors (false negative; β) and the statistical power of a study
- Underpowered studies may miss real effects or produce inflated effect size estimates (Type M error)



Wrap up

The PARACHUTE trial

RESEARCH

Parachute use to prevent death and major trauma when jumping from aircraft: randomized controlled trial

Robert W Yeh,[1] Linda R Valsdottir,[1] Michael W Yeh,[2] Changyu Shen,[1] Daniel B Kramer,[1] Jordan B Strom,[1] Eric A Secemsky,[1] Joanne L Healy,[1] Robert M Domeier,[3] Dhruv S Kazi,[1] Brahmajee K Nallamothu[4] On behalf of the PARACHUTE Investigators

WHAT IS ALREADY KNOWN ON THIS TOPIC

Parachutes are routinely used to prevent death or major traumatic injury among individuals jumping from aircraft, but their efficacy is based primarily on biological plausibility and expert opinion

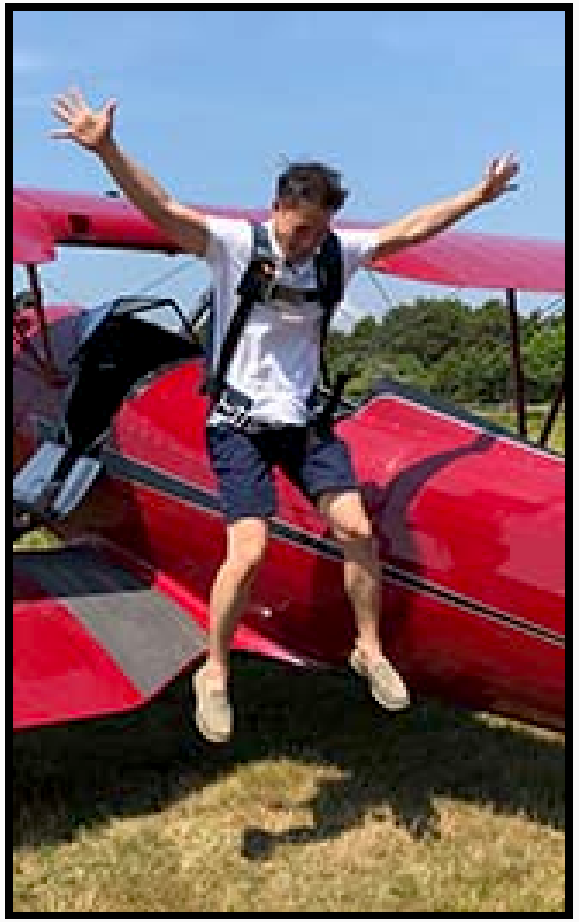
No randomized controlled trials of parachute use have yet been attempted, presumably owing to a lack of equipoise

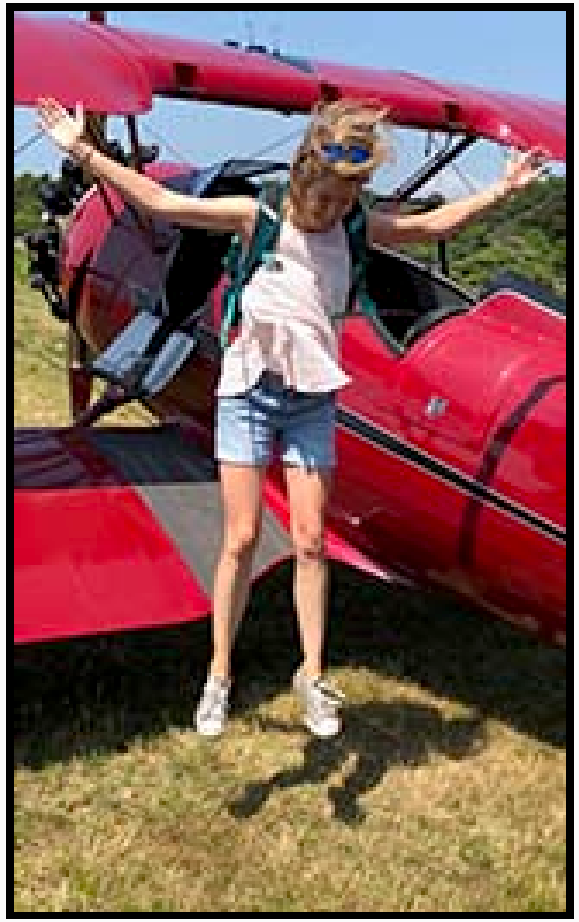
WHAT THIS STUDY ADDS

This randomized trial of parachute use found no reduction in death or major injury compared with individuals jumping from aircraft with an empty backpack

Lack of enrolment of individuals at high risk could have influenced the results of the trial

The PARACHUTE trial





Closing remarks

> *When the designers of scientific studies don't employ statistics with sufficient care, they can sink years of work and thousands of dollars into research that cannot possibly answer the questions it is meant to answer*

A. Reinhart, *Statistics Done Wrong*, 2015, No Starch Press

Closing remarks

> *To consult the statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of.*

R.A. Fisher

Thank you